

Chapter 1

Latent Space Analysis

The ability of a neural network to generalize provides evidence that the model has learned more than a simple memorization of the training examples. Instead, it must construct internal representations that capture regularities present in the data. These representations play a central role in the network’s ability to solve a task. Understanding their structure is therefore a key step toward understanding how neural networks achieve generalization.

In classical machine learning approaches, practitioners had to manually design data representations in order to make them usable by simple models. Deep learning introduces a paradigm shift: rather than relying on hand-crafted features to meaningfully represent data, it automatically learns these representations. Through the successive activation of hidden layers, neural networks gradually construct representations that become increasingly relevant to the task under consideration. As discussed in Chapter ??, these representations are not meaningful at initialization, they emerge progressively during training as the model optimizes its objective. It is precisely through these learned representations that a trained network is able to map an input to an output, including for examples never encountered during training.

The study of these representations has therefore become a central topic in deep learning research, particularly in understanding how information is organized within them, what concepts they encode, and how they influence model behavior. In this chapter, we first make explicit the working hypothesis that underlies most research on latent space analysis. In this manuscript, we refer to this assumption as the “concept hypothesis.” To do so, we introduce the notions of abstraction, concept, latent space, and semantics. Once the notions of this conceptual framework has been established, we review the main methods that aim to identify and extract concepts from latent representations.

1.1 Concept Hypothesis

In this section, we introduce the main working hypothesis underlying much of the literature on latent space analysis. Although rarely stated explicitly, this hypothesis is implicitly present in the recurring use of notions such as abstraction, representation, latent space, concept, and semantics. The goal of this section is to make this conceptual framework explicit by formally defining its main

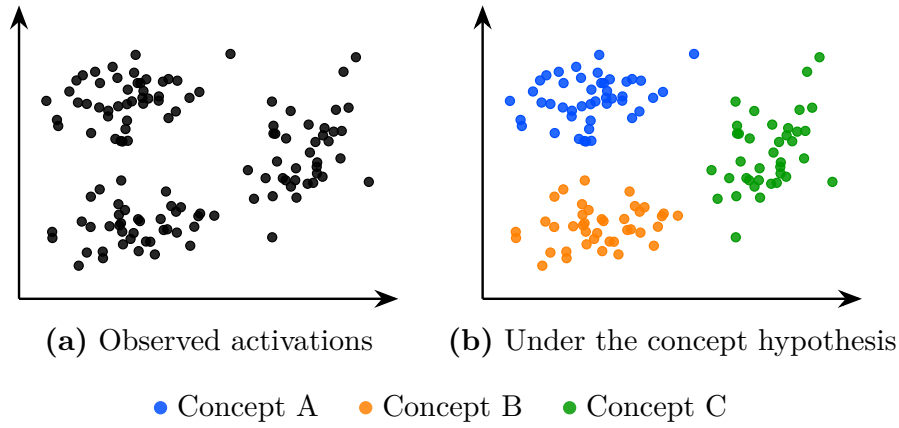


Figure 1.1: The same latent space activations seen without concept hypothesis (a) and under the working hypothesis that concepts emerge in the latent space (b). Colors represent hypothetical concept memberships assigned by the model; they are not observed labels. Under this hypothesis the structure of the latent space becomes meaningful and amenable to analysis.

components.

This hypothesis originates from the need to explain the generalization ability of deep neural networks observed after training, as discussed in Section ??.

Formally, a model is said to generalize when, for an input $x \in \mathcal{X}$ not seen during training, it still produces an output $\hat{y}$ close to the true output y . Such behavior cannot be explained by simple memorization of training samples. Instead, the model must extract and exploit underlying regularities in the target function. This requires a process of abstraction, as defined in Definition 1.1.1, whereby raw inputs are transformed into internal structures that are meaningful for the task.

Definition 1.1.1: Abstraction

Abstraction is the process by which a model transforms an input into a task-relevant internal form. It captures and restructures information in order to represent relationships that are useful for prediction or decision-making.

Through this abstraction process, a neural network does not directly map x to y . Instead, as illustrated in Figure ??, it computes a sequence of intermediate activations across hidden layers. These activations correspond to internal encodings of the input, which we refer to as representations in the sense of Definition 1.1.2.

Definition 1.1.2: Representation

A representation is the internal encoding of an input produced by a trained neural network. It corresponds to the activations of one or several layers and provides a transformed view of the input.

Each hidden layer refines the representation of the input. It progressively

53 transforms it into a structure more directly aligned with the output space. This
 54 process can be described as a sequence of transformations where each $h^{(l)}$ is a
 55 representation:

$$x \rightarrow h^{(1)} \rightarrow h^{(2)} \rightarrow \dots \rightarrow h^{(L-1)} \rightarrow y$$

56 The set of all such representations produced by a layer or a model over a
 57 dataset defines its latent space, as defined in Definition 1.1.3.

Definition 1.1.3: Latent Space

The latent space is the set of all representations produced by a layer or model over a dataset. It is shaped by learning and organizes representations according to task-relevant properties.

58
 59 The existence of meaningful structure in the latent space is closely related
 60 to generalization. Under the concept hypothesis, we assume that the neural
 61 network internally forms task-dependent abstractions, hereafter referred to as
 62 concepts and formally defined in Definition 1.1.4. These concepts are not
 63 directly observable, but are expected to manifest in the latent space as recurring
 64 and structured patterns in the learned representations. Under this view, a
 65 representation can be interpreted as a composition of multiple concepts, each
 66 contributing to the model's prediction.

Definition 1.1.4: Concept

A concept is a task-dependent abstraction that captures regularities in the input space relevant for predicting the target output.

67
 68 Another property that can be considered when analyzing concepts is their
 69 relationship to human interpretation. This relationship is captured by the notion
 70 of semantics, defined in Definition 1.1.5. Under this definition, concepts may
 71 exhibit varying degrees of semantic content. Some concepts can be readily
 72 associated with human-interpretable notions, whereas others may contribute to
 73 task performance while remaining difficult to interpret.

Definition 1.1.5: Semantics

Semantics refers to the alignment between concepts in the latent space and concepts used by a human observer. A structure is semantic if it can be consistently associated with a human-interpretable concept.

74
 75 It is important to note that this conceptual framework is not yet supported
 76 by a complete theoretical understanding of deep neural networks. Current
 77 mathematical tools do not provide a rigorous characterization of their internal
 78 representations and do not formally prove that concepts emerge as identifiable
 79 structures within latent spaces. Nevertheless, this perspective constitutes a
 80 widely adopted working hypothesis in the field of representation and latent space
 81 analysis.

82 The remainder of this chapter is devoted to reviewing methods for analyzing
 83 latent spaces and extracting the concepts they encode.

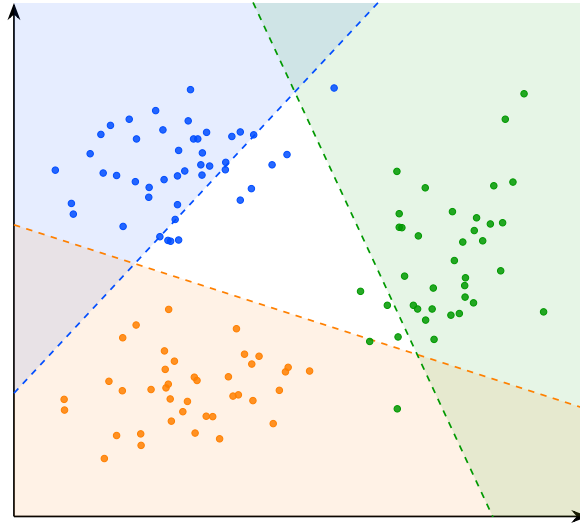


Figure 1.2: Decision boundaries of the linear probes for concepts A, B, and C. The colored regions indicate the areas classified positively by each probe.

1.2 Probing

Probing methods aim to determine whether a given human-defined concept is encoded within the internal representations of a neural network. More specifically, they investigate whether the presence or absence of a selected concept can be inferred from embeddings. The underlying assumption is that if a concept can be reliably decoded from these representations, it is likely that the network encodes information related to it in order to perform its task.

A probing analysis begins by identifying a set of interpretable concepts that the network could potentially use to accomplish its task. Each example in the dataset is then annotated to indicate the presence or absence of the studied concept. The data are subsequently projected into the latent space of the pretrained model. From these latent representations, a classifier is trained in a supervised manner to predict the presence of the considered concept. This classifier is referred to as a probe, since its role is to determine whether a concept is present or not. The role of this probe is therefore to assess whether the information associated with the concept is accessible from the network's internal representations.

The classifier used to detect the presence or absence of a concept must remain intentionally simple. Indeed, if the probe has too much capacity, it may reconstruct the target information from complex correlations that do not necessarily correspond to a representation explicitly encoded in the latent space. In such a case, it becomes difficult to determine whether the concept is genuinely represented by the network or merely reconstructed through the expressive power of the classifier. For this reason, probing methods often rely on linear classifiers. A linear separation suggests that the information is easily accessible to the neural network and potentially usable during decision making.

If the classifier achieves performance significantly above chance level, this suggests that the latent space contains exploitable information related to the

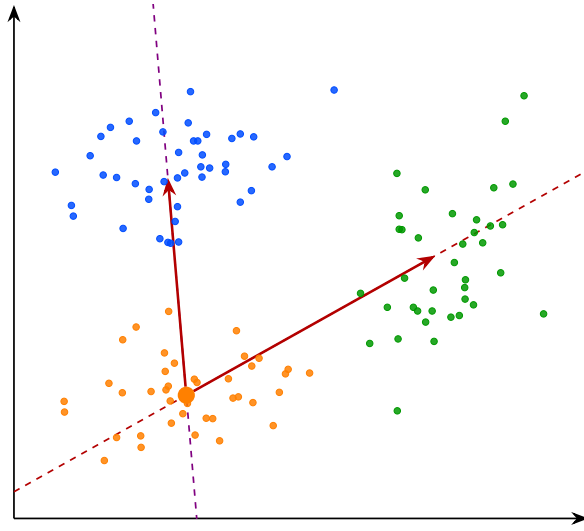


Figure 1.3: Directions interprétables dans l'espace latent reliant le cluster B aux clusters A et C. Les lignes pointillées représentent les directions globales, tandis que les flèches indiquent les vecteurs entre centroïdes.

112 considered concept. However, the fact that a concept can be decoded from latent
 113 representations does not necessarily imply that the model actually uses this
 114 information to perform its task.

115 To ensure that representations play a causal role in the model's output, it is
 116 possible to perform interventions on the latent representations. An intervention
 117 consists of modifying a latent representation in a controlled manner by activating
 118 or deactivating a given concept, and then studying its impact on the model's
 119 output. If the resulting change in the model's output is consistent with the
 120 applied perturbation, this suggests that the representation had a causal role.

121 The main limitation of this approach is that the concepts under investigation
 122 must be defined in advance before any probing analysis can be performed. This
 123 introduces a strong bias in what can be studied. Moreover, annotating a dataset
 124 for each concept is extremely time-consuming and requires significant human
 125 effort. Interventions in latent space are also difficult to interpret, as it is not
 126 always clear what constitutes a meaningful change in the network's output under
 127 a given perturbation.

128 This set of limitations makes probing a highly human-centric approach,
 129 strongly dependent on the choice of investigated concepts and on the interpreta-
 130 tion of interventions. To overcome this bias, other methods for studying latent
 131 spaces have been developed, particularly those aiming to discover concepts in an
 132 unsupervised manner.

133 1.3 Interpretable Directions

134 In latent space analysis, probing methods are generally limited to a relatively
 135 simple task: predicting the presence or absence of a given concept from a latent
 136 representation. This approach makes it possible to assess whether information is

137 encoded in the latent space, but it does not provide a detailed characterization
138 of its structure.

139 A more advanced line of investigation focuses on directions in latent space.
140 Let $\mathbf{z} \in \mathbb{R}^d$ denote a latent representation. Empirically, in certain models, moving
141 a latent representation along a direction $\mathbf{v} \in \mathbb{R}^d$ results in coherent changes in
142 the model’s output.

143 More precisely, consider the perturbed representation

$$\mathbf{z}' = \mathbf{z} + \alpha \mathbf{v},$$

144 where $\alpha \in \mathbb{R}$ controls the magnitude of the intervention. The resulting change in
145 output often varies smoothly with α , suggesting an approximately linear structure
146 in latent space with respect to certain semantic factors. This observation has
147 led to the hypothesis that neural networks may organize representations in a
148 partially linear manner with respect to semantic concepts.

149 Such vectors $\mathbf{v}$ are referred to as interpretable directions, since they induce
150 transformations that can be associated with semantic meaning. The goal of
151 latent space analysis through interpretable directions is therefore to identify
152 these semantically meaningful axes of variation.

153 This phenomenon has been extensively studied in generative image models. In
154 this context, controlled perturbations of latent representations along interpretable
155 directions can induce specific semantic transformations, such as object rotation,
156 scaling, or age modification in facial images. These observations have led to the
157 development of the field of latent space editing, whose objective is to manipulate
158 generated content directly through modifications in latent space rather than
159 through changes in the input prompt.

160 However, a central challenge remains: identifying such interpretable directions
161 is far from trivial. Most existing approaches rely on auxiliary models that encode
162 human-defined semantics, such as CLIP, to relate latent variations to textual
163 concepts. Consequently, these methods inherit a limitation similar to that of
164 probing techniques, since concept discovery depends on a pre-existing semantic
165 framework defined by humans.

166 To overcome this limitation, a research direction known as unsupervised
167 semantic discovery has emerged. Its goal is to uncover interpretable directions
168 in latent spaces without explicit semantic supervision. In these approaches, one
169 model applies perturbations along random directions in latent space, while a
170 second model attempts to predict or recognize the applied transformation. If
171 a stable correspondence emerges between the perturbation and its prediction,
172 this suggests that the considered direction corresponds to a coherent structure
173 in latent space.

174 In this framework, human intervention occurs only a posteriori, to verify
175 whether the discovered directions indeed correspond to interpretable semantic
176 variations. This approach enables the capture of representations that go beyond
177 the mere presence or absence of a concept, including notions such as the intensity
178 or degree of an attribute. As such, it follows a philosophy similar to causal
179 probing, while also enabling explicit interventions to assess the causal role of
180 latent representations.

181 However, despite these advances, several challenges remain.

182 First, it appears that not all concepts encoded by neural networks are linearly
183 identifiable in latent space. While some directions correspond to well-defined
184 semantic factors, many others do not admit such a simple linear structure.

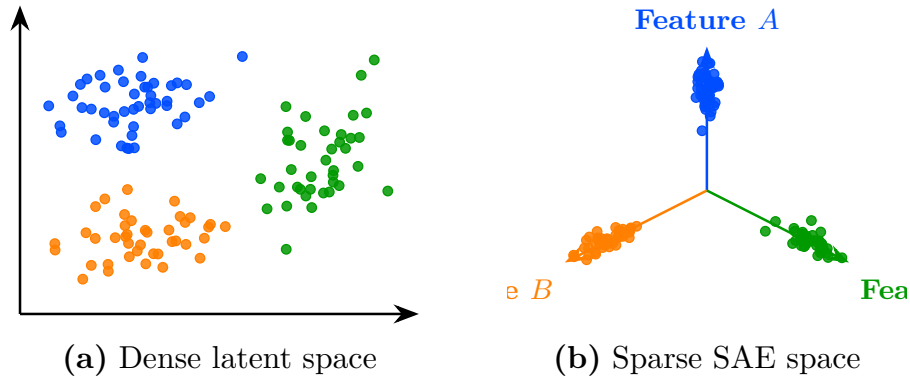


Figure 1.4: Transformation de l'espace latent dense vers une représentation sparse de type SAE. À gauche (a), les embeddings sont entremêlés dans l'espace latent initial. À droite (b), la représentation sparse aligne chaque concept sur un axe de feature dédié.

In addition, latent representations often exhibit a phenomenon known as entanglement. Let $f : \mathbb{R}^d \rightarrow \mathbb{R}^k$ denote a mapping from latent space to a set of semantic attributes. Ideally, one would like to associate each semantic factor $i \in \{1, \dots, k\}$ with a direction $\mathbf{v}_i \in \mathbb{R}^d$ such that:

$$f(\mathbf{z} + \alpha \mathbf{v}_i)_j \approx f(\mathbf{z})_j \quad \text{for } j \neq i,$$

and

$$f(\mathbf{z} + \alpha \mathbf{v}_i)_i \text{ varies significantly with } \alpha.$$

In practice, this property does not hold strictly. Instead, a single direction $\mathbf{v}$ often induces changes in multiple components of $f(\mathbf{z})$, i.e.:

$$\frac{\partial f_j(\mathbf{z})}{\partial \alpha} \neq 0 \quad \text{for several } j.$$

This phenomenon is referred to as entanglement, indicating that semantic attributes are not cleanly separable in latent space.

This difficulty in isolating independent semantic factors, together with the frequent presence of entangled representations, has motivated a more ambitious hypothesis: the superposition hypothesis.

1.4 Sparse Autoencoder

As discussed previously, interpretability approaches based on latent directions face inherent limitations. Empirically, only a restricted subset of concepts appears to admit a linear representation, and even these directions are often affected by significant entanglement with other concepts.

To account for this behavior, a complementary hypothesis has been introduced, extending the assumption of linear concept encoding in neural networks. It posits that semantic concepts are preferentially represented in a linear fashion, with each concept ideally corresponding to a distinct direction in latent space.

In this idealized setting, semantic attributes would form an approximately orthogonal basis, where each direction independently controls a single factor of variation.

We first consider an idealized representation of the latent space in which semantic concepts are perfectly disentangled. Let $\mathbf{h} \in \mathbb{R}^d$ denote a latent embedding. In this setting, each concept is associated with a unique direction $\mathbf{v}_i \in \mathbb{R}^d$, and the representation of a given input can be expressed as a linear combination of these concept directions:

$$\mathbf{h} = \sum_{i=1}^m a_i \mathbf{v}_i,$$

where $a_i \in \mathbb{R}$ represents the activation of concept i .

In the ideal case, the set of concept directions is assumed to be approximately orthogonal, such that each direction independently controls a single semantic factor. Moreover, for a given input, only a small subset of coefficients a_i is non-zero, so that each embedding corresponds to a sparse combination of a few active concepts.

In this idealized setting, concept directions are assumed to form a structured basis of the latent space, in which each concept is independently controllable.

When the representational capacity of the model is limited, such a decomposition leads to interference between concepts. In this regime, multiple semantic concepts are encoded within shared directions, a phenomenon known as superposition. This can be interpreted as a consequence of compression pressure, where the model prioritizes efficient encoding over strict factorization of concepts.

This perspective is closely related to recent work in mechanistic interpretability, where neural representations are assumed to decompose into sparse, approximately linear concepts. It also connects naturally to classical results in dictionary learning and sparse coding, where signals are represented as sparse combinations of elements from an overcomplete basis.

Sparse autoencoders instantiate this idea in practice by learning an overcomplete dictionary of concepts while enforcing sparsity constraints on activations. From this perspective, they can be interpreted as a tool for recovering a more disentangled basis of concepts from a superposed representation space.

Despite their empirical success, sparse autoencoders rely on strong structural assumptions that are not always justified. First, the learned concepts are not guaranteed to be identifiable: different training runs may yield different decompositions of the same latent space, even when the reconstruction performance is similar. Second, the assumption that the underlying representation is approximately linearly decomposable may not hold uniformly across all regions of the latent space, especially in highly nonlinear regimes. Finally, sparsity itself introduces a modeling bias that may favor certain types of concepts while suppressing others.

These limitations suggest that sparse autoencoders, while powerful, do not fully resolve the problem of entanglement in neural representations. They instead provide one possible coordinate system in which parts of the structure become more explicit, but not necessarily a complete or canonical one.

From this perspective, both interpretable directions and sparse autoencoders rely on a shared underlying assumption: that semantic structure can be recovered

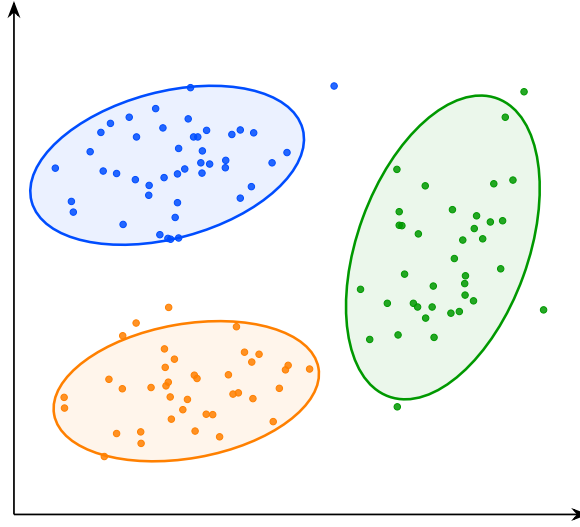


Figure 1.5: Visualisation de la structure de l'espace latent. Les points représentent les embeddings associés aux concepts A, B et C. Les ellipses indiquent l'étendue approximative de chaque groupe dans l'espace latent et mettent en évidence leur séparation géométrique.

through linear or approximately linear decompositions. While this assumption has led to significant empirical progress, it remains theoretically fragile.

The following section therefore introduces a complementary approach that relaxes this constraint and does not require a globally linear feature decomposition of the latent space.

1.5 Clustering

The latent space analysis methods presented so far all rely, either explicitly or implicitly, on assumptions about the linear organization of concepts within the latent space. For probing methods, concepts must be at least partially linearly separable for simple probes to succeed. Interpretability through latent directions assumes that variations associated with a concept can be captured by moving along a linear axis of the latent space. Similarly, the superposition hypothesis assumes that concepts are represented through linear directions that become entangled because of representational capacity constraints. Although these assumptions have led to numerous empirical successes, their theoretical justification remains limited. More importantly, several observations suggest that the linearity hypothesis may not fully capture the structure of latent representations.

For instance, some concepts can only be recovered using non-linear probes. Likewise, methods based on interpretable directions typically identify only a limited number of semantic factors. Finally, in sparse autoencoder approaches, enforcing excessive sparsity often degrades reconstruction quality and interpretability, suggesting that the underlying representation may not be perfectly described by a sparse linear decomposition.

Taken together, these observations indicate that semantic concepts may not always be organized according to a simple linear structure. This motivates the exploration of alternative approaches that rely on weaker assumptions regarding the geometry of latent spaces. One such approach is clustering-based analysis. Rather than assuming that concepts correspond to directions or linear subspaces, clustering methods only assume that semantically similar inputs produce nearby latent representations. Conversely, inputs that differ significantly from a semantic perspective are expected to be located farther apart in the latent space.

Under this assumption, the discovery of concepts can be reformulated as the identification of groups of embeddings that occupy the same region of the latent space. Concepts are therefore no longer associated with directions, but with clusters of neighboring representations. This idea forms the basis of the research field known as deep clustering. In these approaches, a collection of inputs is first projected into a latent space using a neural network. The resulting embeddings form a point cloud on which standard clustering algorithms can be applied. The obtained clusters are then interpreted as representing distinct semantic concepts.

This methodology enables the discovery of semantic structure in the latent space without requiring concepts to be linearly encoded. As a result, it can reveal forms of organization that would remain inaccessible to methods based solely on linear separability or interpretable directions. Deep clustering has been particularly successful in unsupervised image classification. Typically, an autoencoder is trained to learn a compact latent representation that preserves the essential information contained in the input data. Clustering algorithms are then applied to the learned embeddings. Empirically, the resulting clusters often correspond to meaningful semantic categories, despite the absence of supervision during training.

More advanced approaches combine representation learning and clustering objectives within a single optimization process. In this setting, additional loss terms encourage embeddings to organize themselves around cluster centroids during training. This often produces latent spaces that are easier to analyze and improves the quality of the discovered semantic structure.

Despite its effectiveness, clustering-based analysis also presents important limitations. First, as with most unsupervised semantic discovery techniques, the interpretation of the resulting clusters remains partially subjective. While benchmark datasets often provide class labels that facilitate evaluation, there is generally no guarantee that the discovered clusters correspond to the categories that humans consider most relevant. Alternative semantic groupings may be equally valid.

Second, clustering methods are highly sensitive to design choices. The number of clusters, the distance metric, and the clustering algorithm itself can significantly influence the results. Selecting these parameters is often challenging and may require substantial empirical tuning. Finally, semantic concepts do not necessarily form well-separated clusters in latent space. Some concepts may overlap, exhibit hierarchical relationships, or vary continuously rather than discretely. In such cases, a clustering-based description may provide only a partial view of the underlying representational structure.